

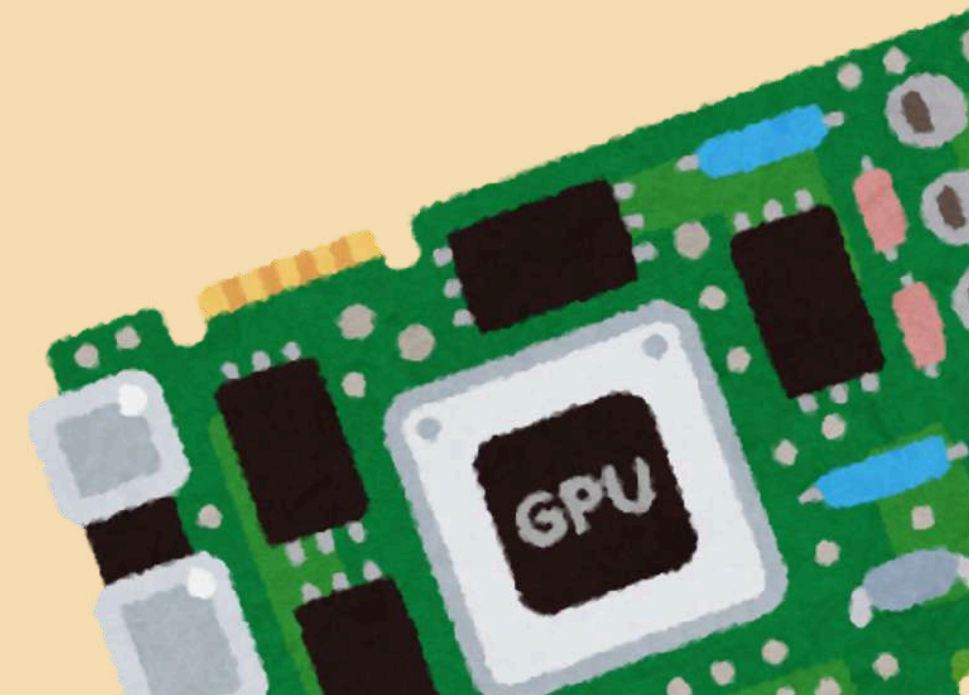
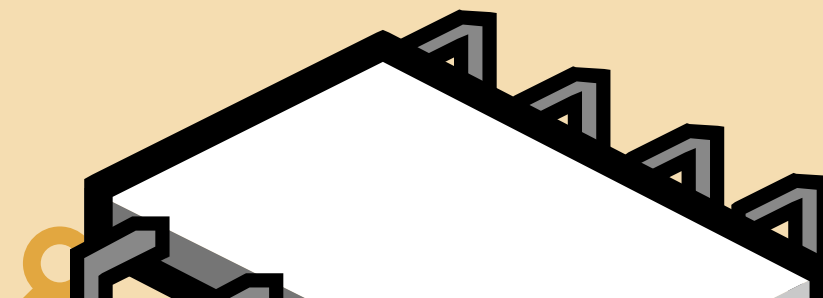
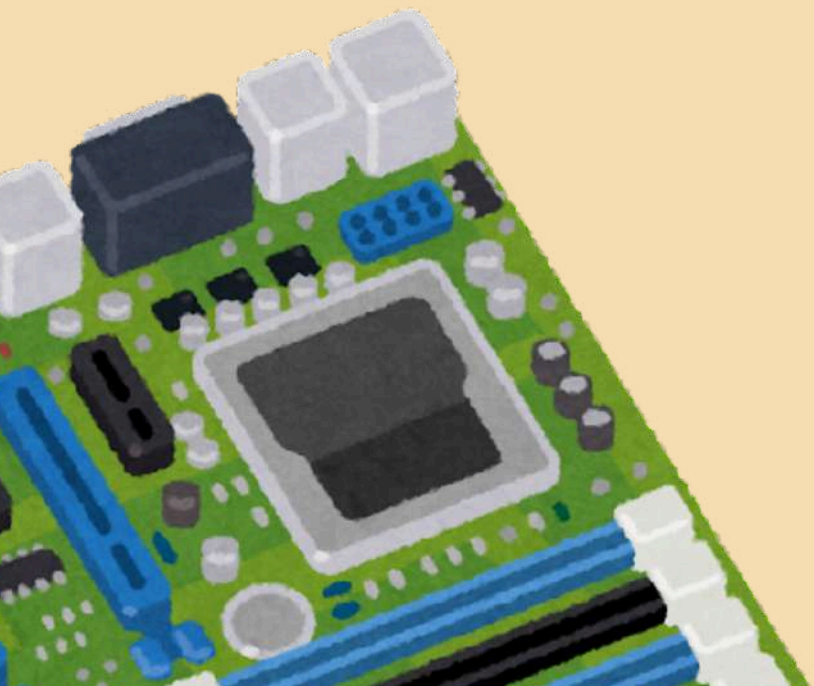
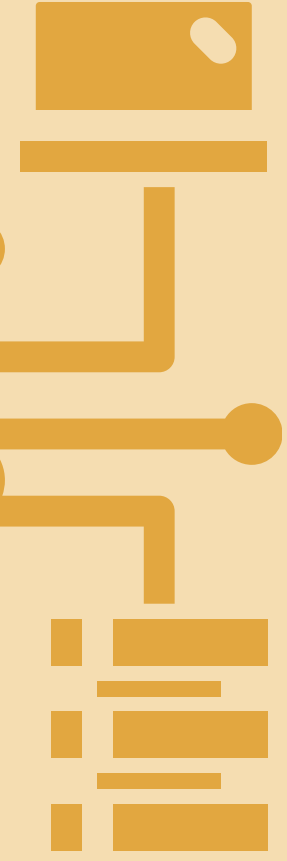
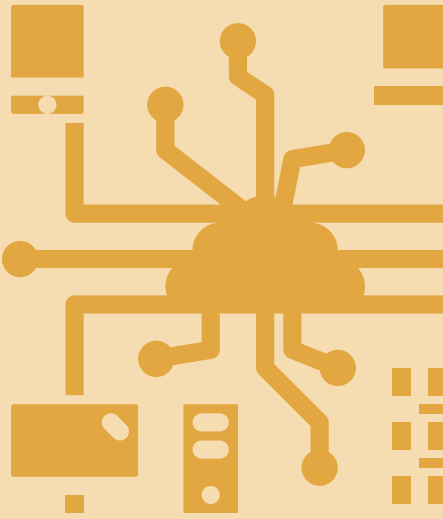
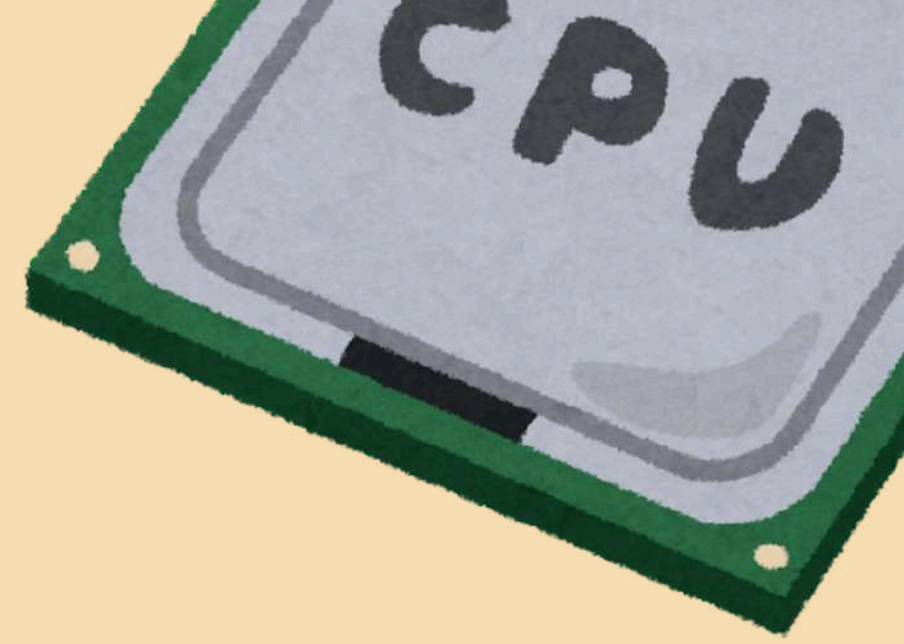


جامعة الإمام عبد الرحمن بن فيصل
IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY
كلية العلوم والدراسات الانسانية – الجبيل

Driver Drowsiness Alarm System

Digital Logic Design Project

Group 1



Our Team

Ghala Alzamanan

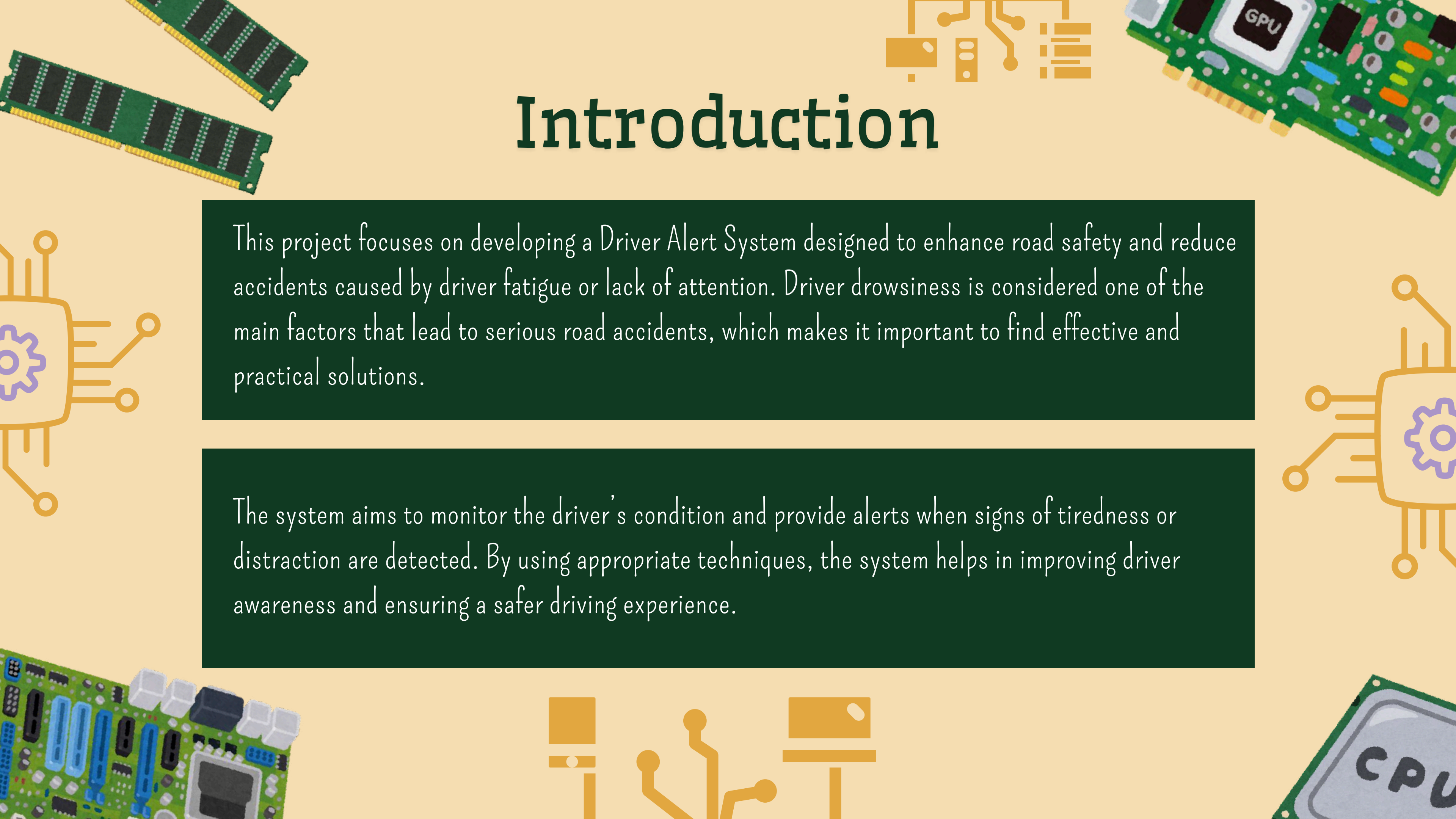
Judi Elahi

Lulwah Albassam

Sara Sultan

Sadeem ALqahtani

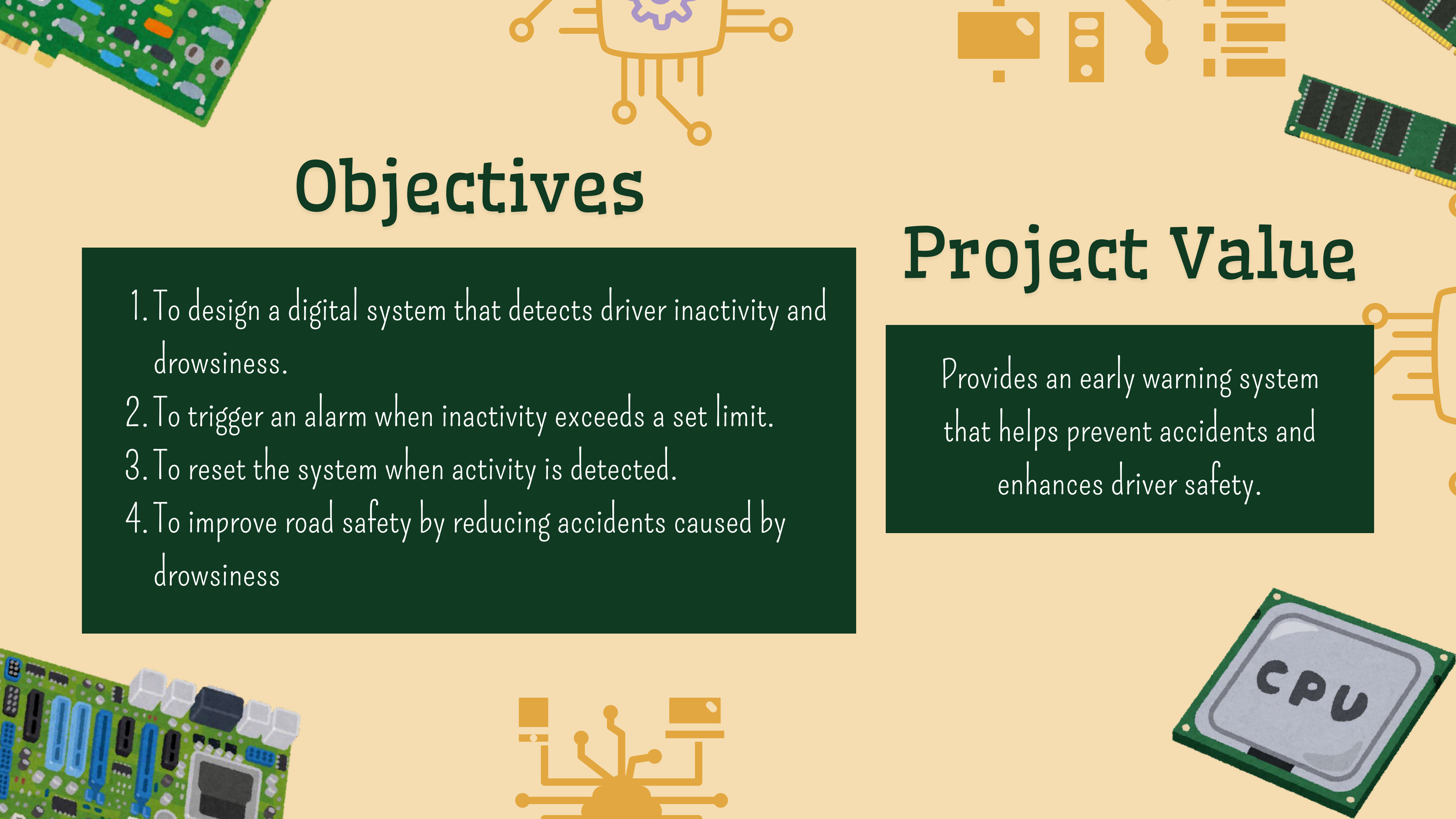
Lamy Alroqi



Introduction

This project focuses on developing a Driver Alert System designed to enhance road safety and reduce accidents caused by driver fatigue or lack of attention. Driver drowsiness is considered one of the main factors that lead to serious road accidents, which makes it important to find effective and practical solutions.

The system aims to monitor the driver's condition and provide alerts when signs of tiredness or distraction are detected. By using appropriate techniques, the system helps in improving driver awareness and ensuring a safer driving experience.

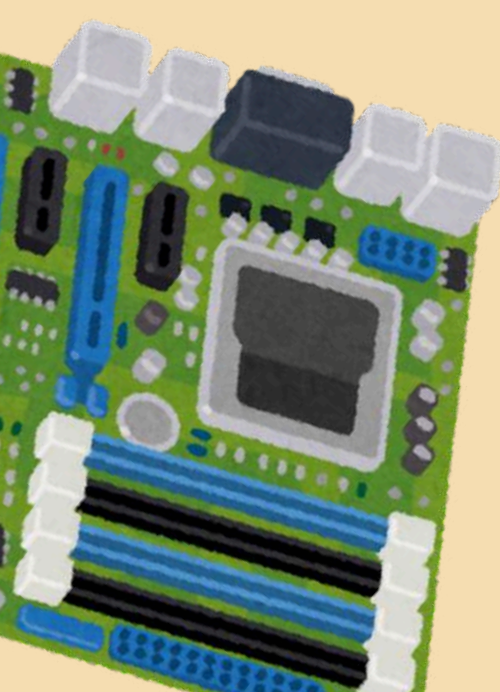
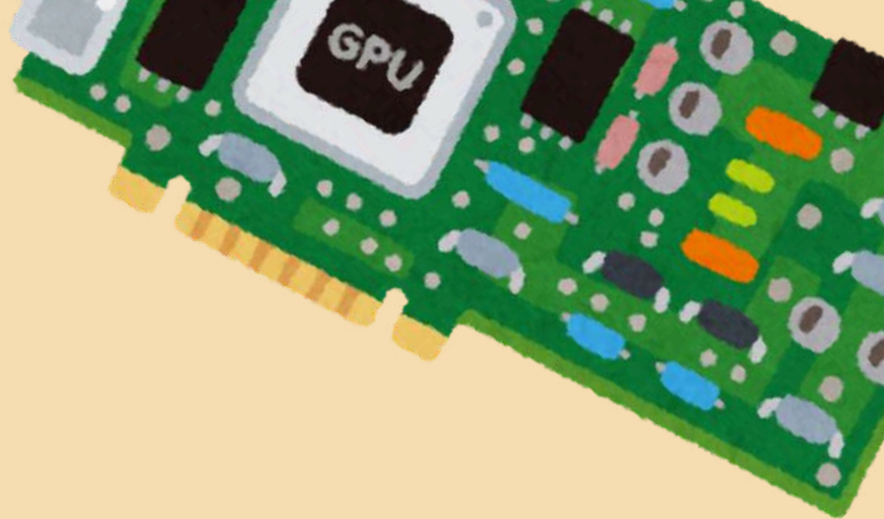
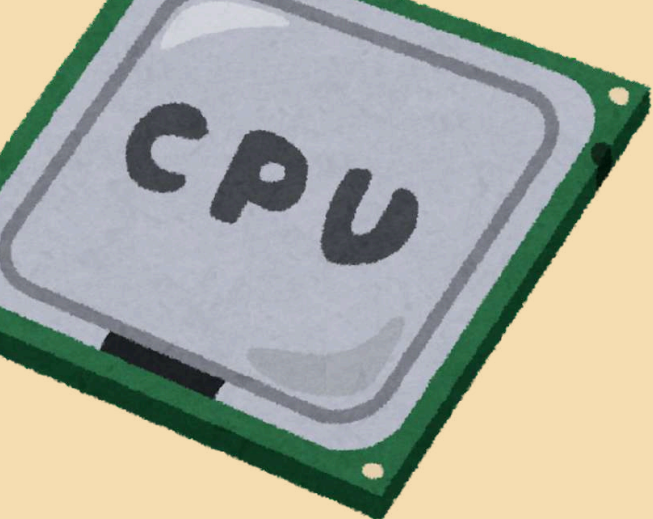


Objectives

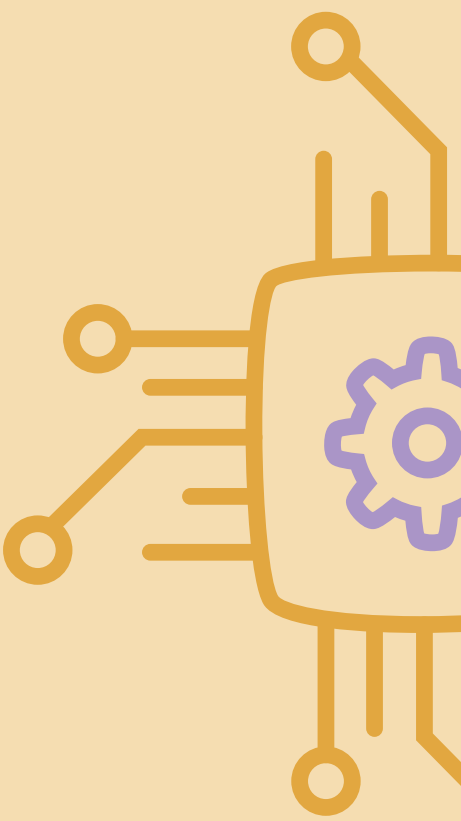
1. To design a digital system that detects driver inactivity and drowsiness.
2. To trigger an alarm when inactivity exceeds a set limit.
3. To reset the system when activity is detected.
4. To improve road safety by reducing accidents caused by drowsiness

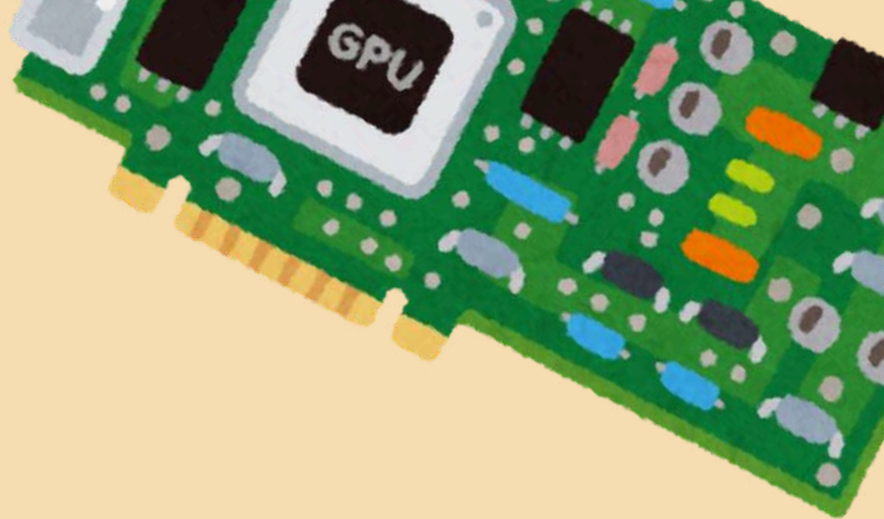
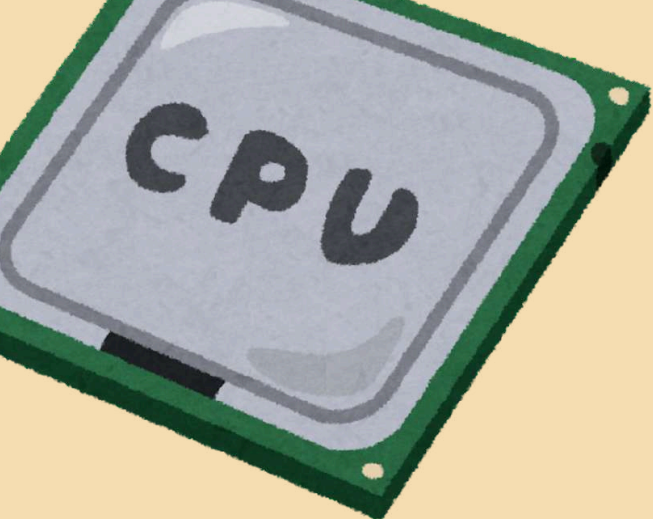
Project Value

Provides an early warning system that helps prevent accidents and enhances driver safety.



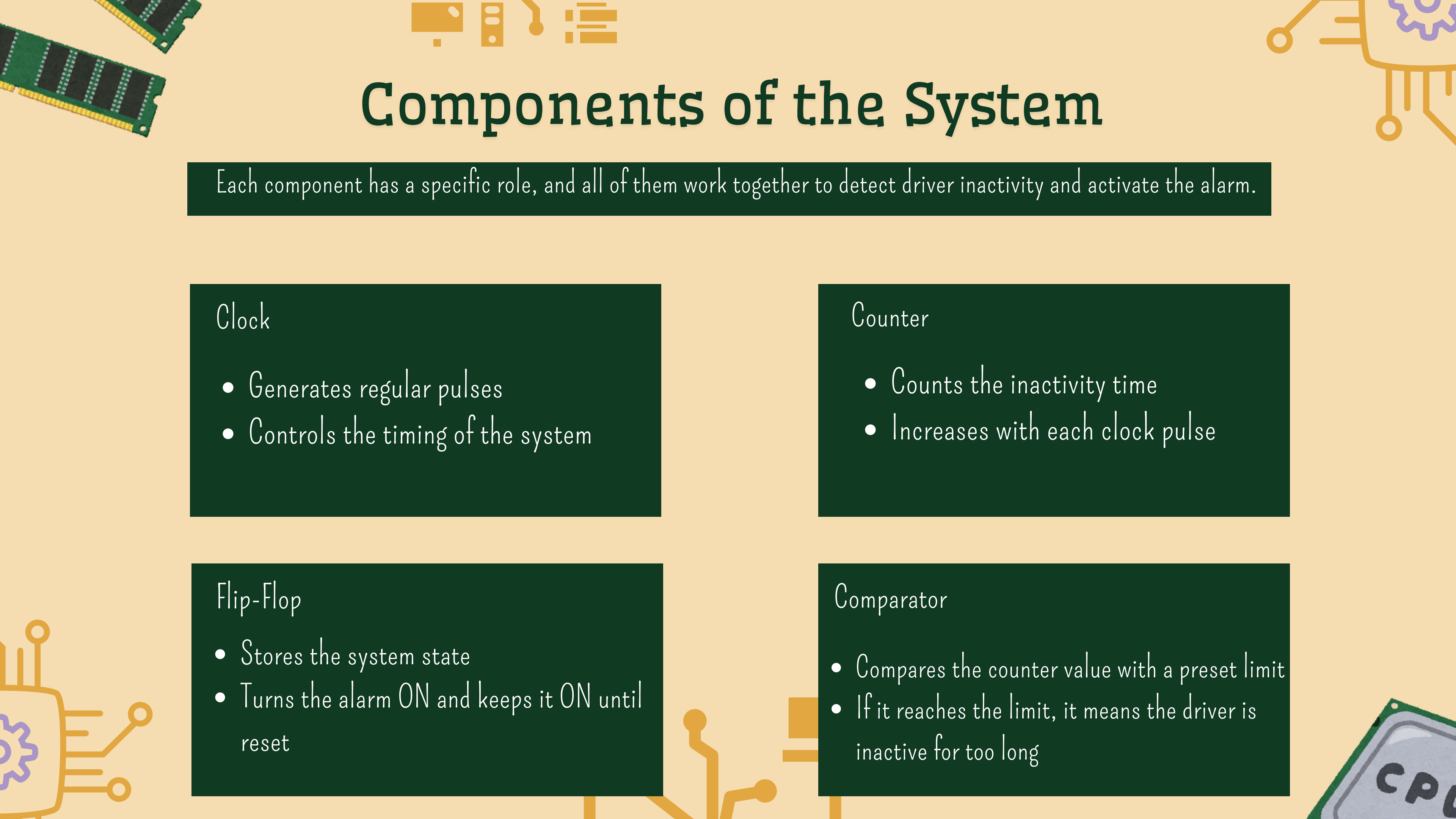
System Idea

- The system monitors the driver's activity
 - It uses a clock and a counter to measure inactivity time
 - The system checks how long the driver is not moving
- 



System Idea

- If inactivity exceeds a specific limit → alarm ON
- If the driver becomes active → system resets
- Example:
 - No movement → alarm ON
 - Movement detected → alarm OFF



Components of the System

Each component has a specific role, and all of them work together to detect driver inactivity and activate the alarm.

Clock

- Generates regular pulses
- Controls the timing of the system

Counter

- Counts the inactivity time
- Increases with each clock pulse

Flip-Flop

- Stores the system state
- Turns the alarm ON and keeps it ON until reset

Comparator

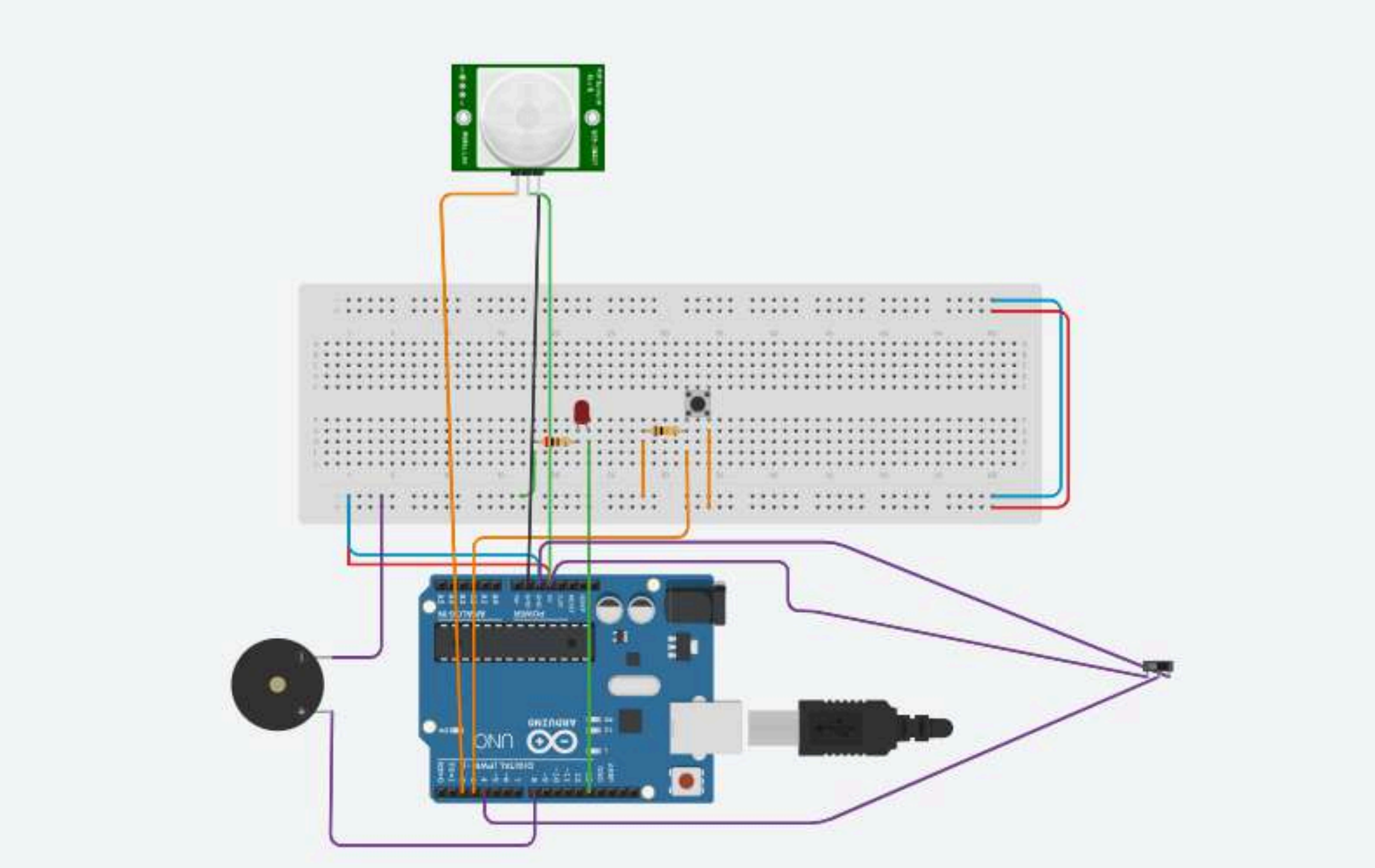
- Compares the counter value with a preset limit
- If it reaches the limit, it means the driver is inactive for too long

Working Mechanism

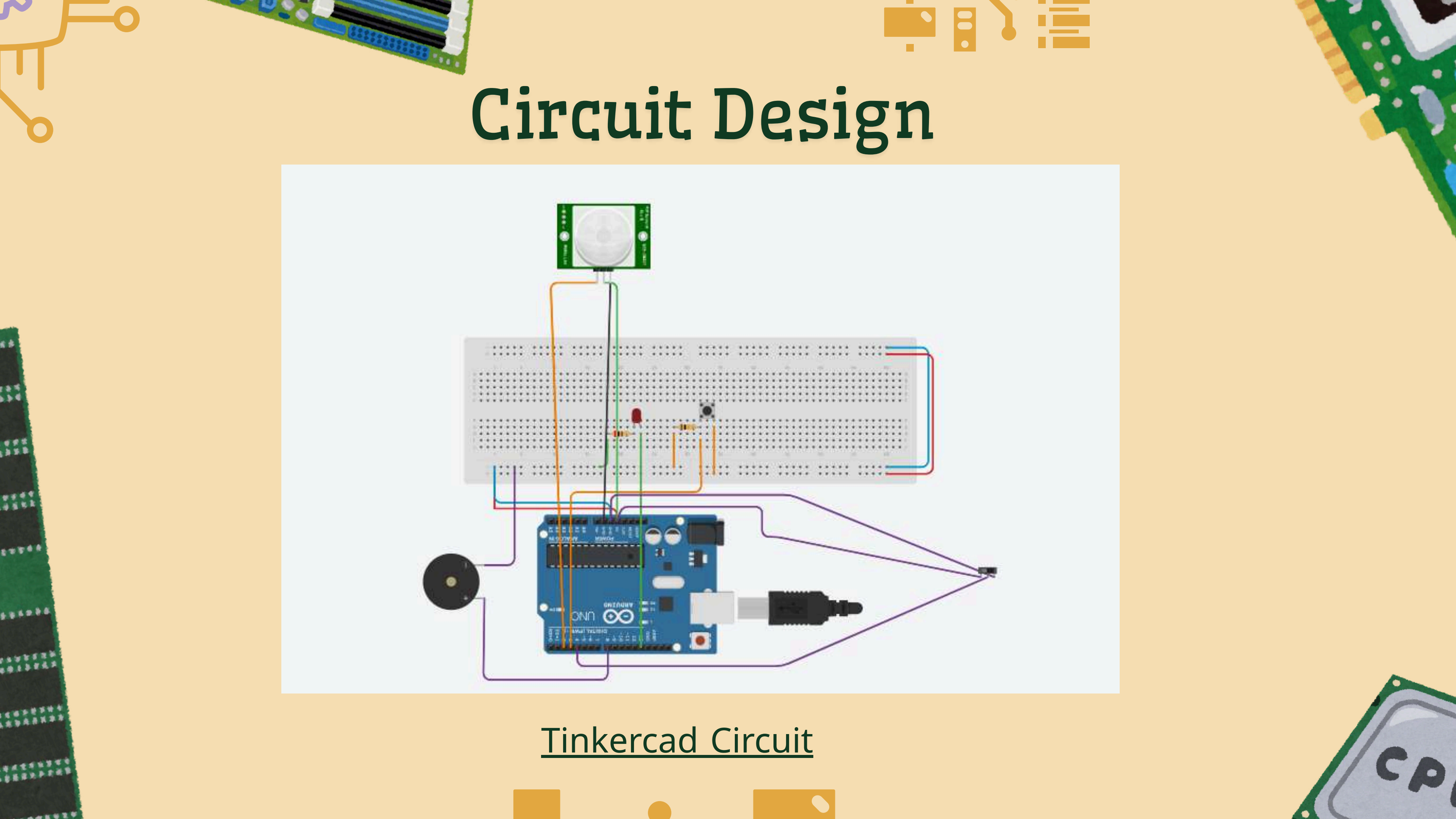
- The PIR sensor detects driver movement.
- If no movement is detected, the timer starts counting.
- When inactivity exceeds the limit, the alarm activates.

- The LED and buzzer warn the driver.
- Pressing the button resets the system.

Circuit Design



Tinkercad Circuit



Circuit Design

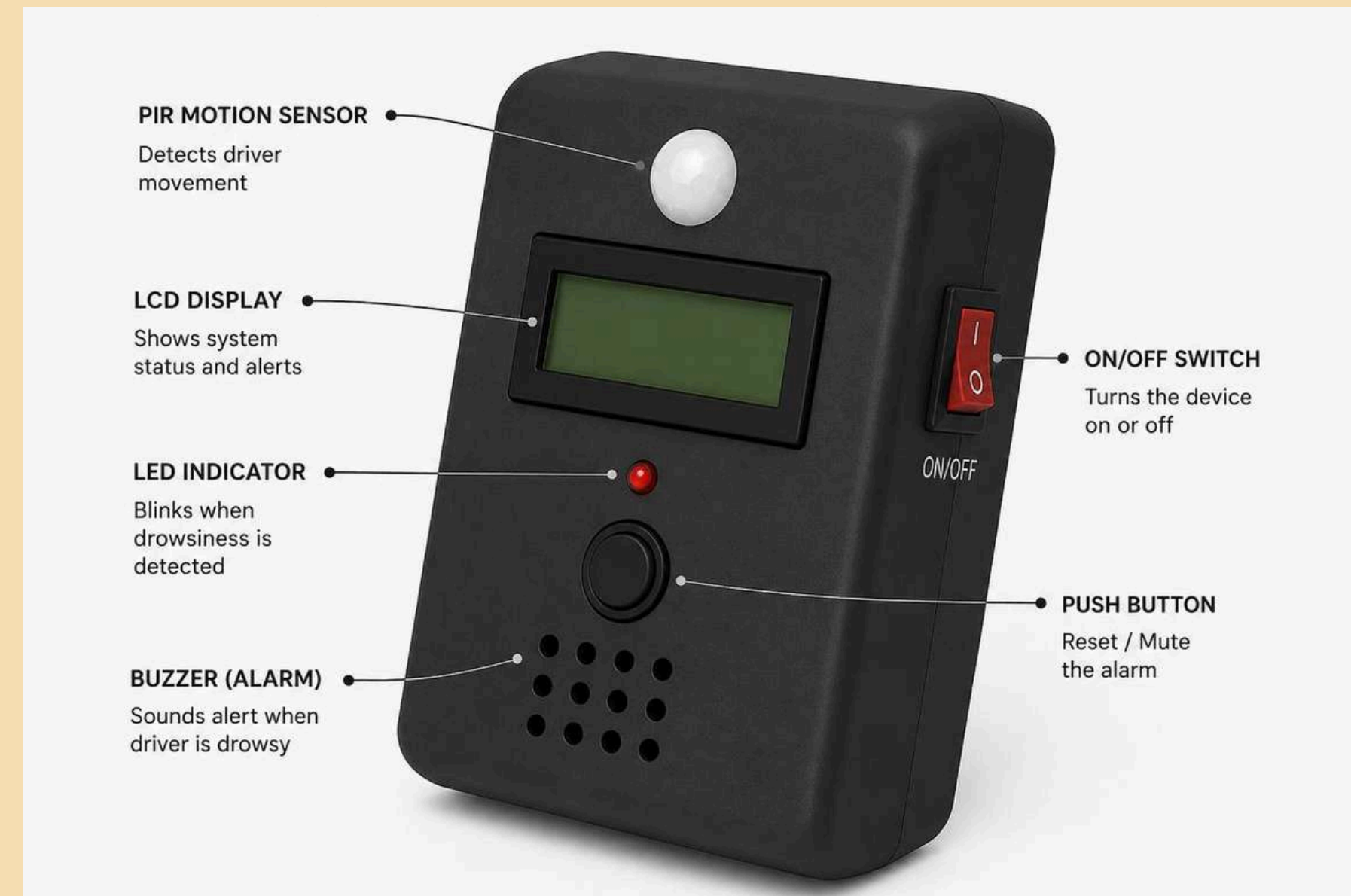
The diagram illustrates a Tinkercad circuit design. It features an Arduino Uno microcontroller board connected to a breadboard. On the breadboard, there is a 555 timer IC, a 10k resistor, and a 100k resistor. A piezo buzzer is also connected to the breadboard. The circuit is powered by a 5V USB connection. The connections are as follows: the 5V pin of the Arduino is connected to the VCC pin of the 555 timer and the positive terminal of the piezo buzzer. The GND pin of the Arduino is connected to the GND pin of the 555 timer and the negative terminal of the piezo buzzer. The output pin of the 555 timer is connected to the positive terminal of the piezo buzzer. The 10k resistor is connected between the output pin of the 555 timer and the positive terminal of the piezo buzzer. The 100k resistor is connected between the output pin of the 555 timer and the GND pin of the Arduino.

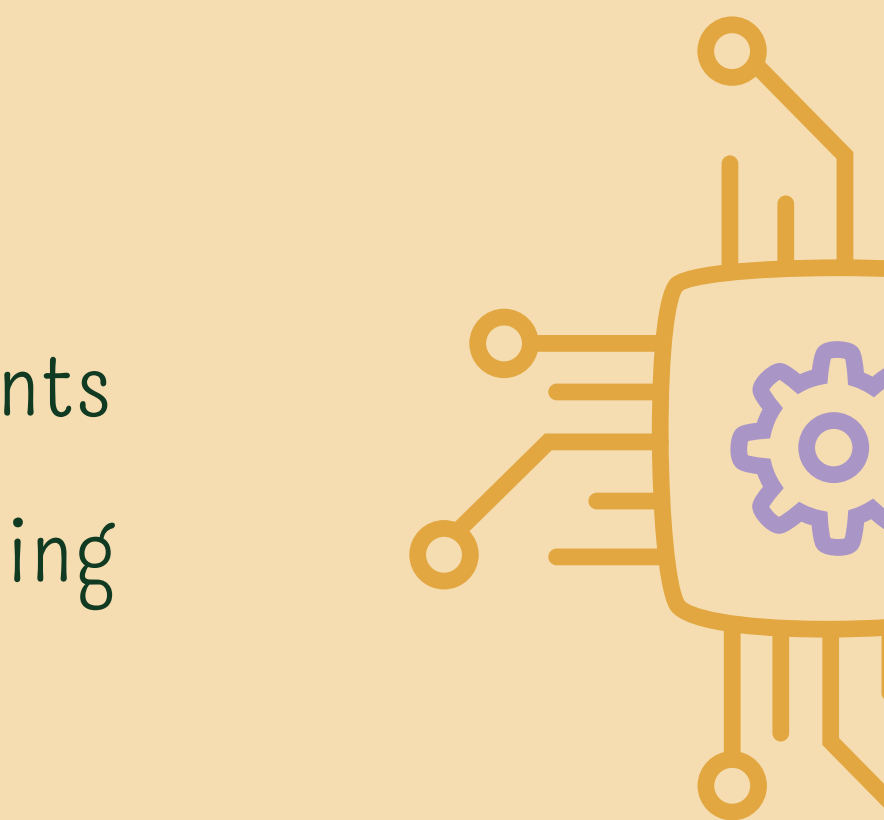
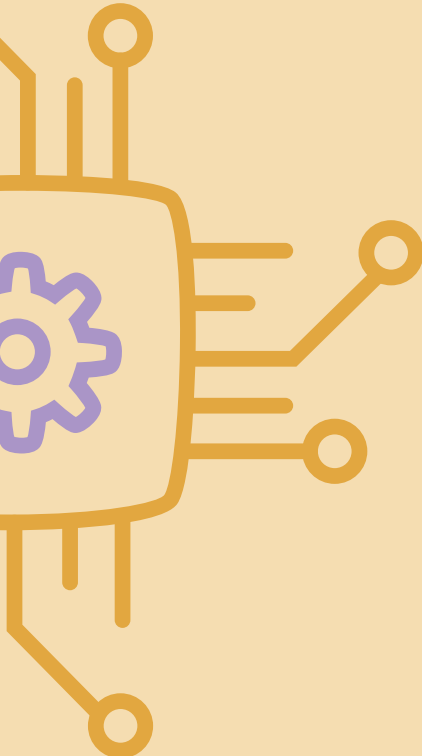
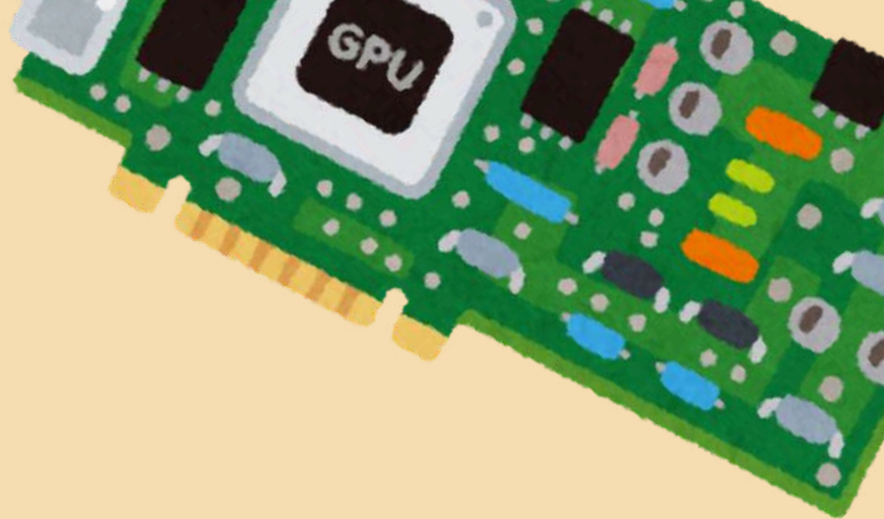
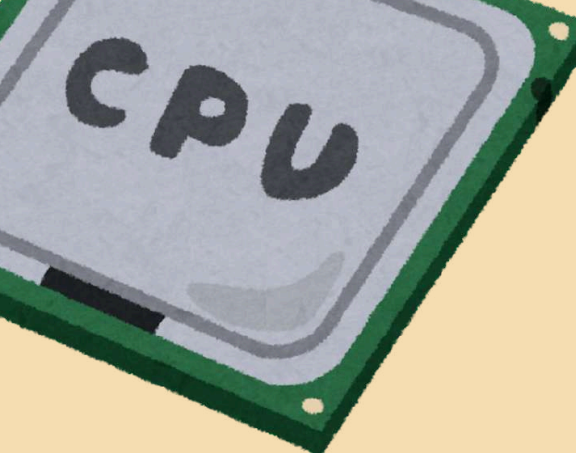
Tinkercad Circuit

3D Product Design

Future Improvements :

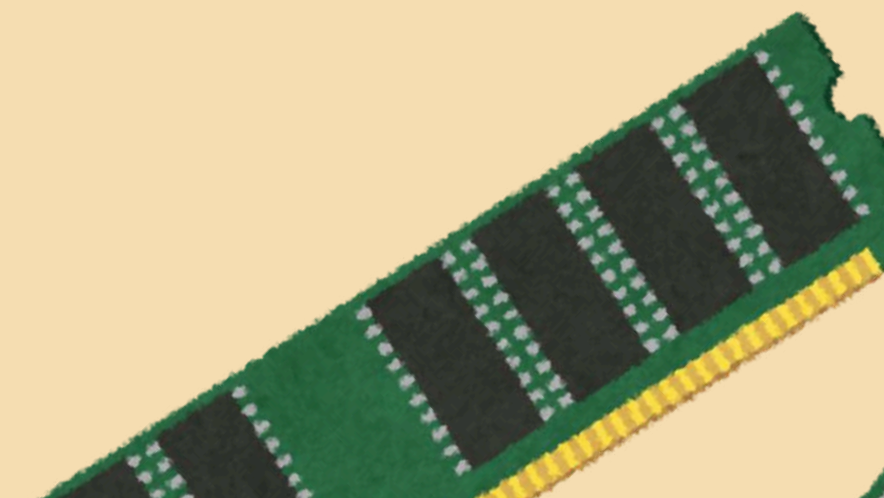
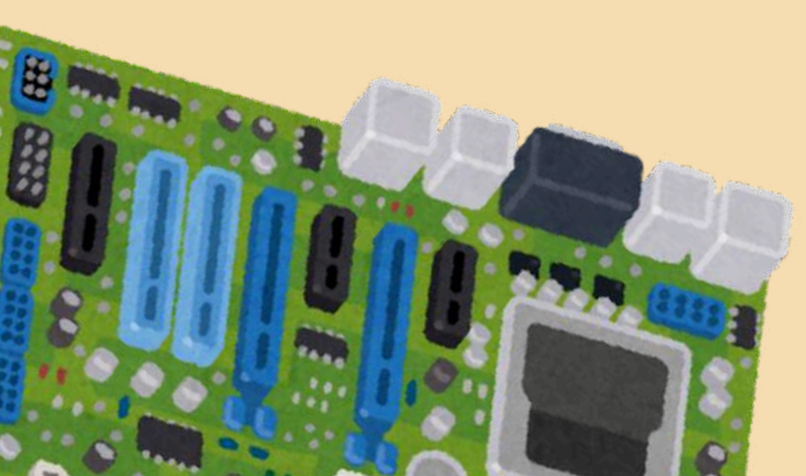
- Add real LCD status display
- Add camera-based eye detection
- Connect the system to vehicle controls
- Add mobile notifications





Conclusion

DriveSafe is a smart safety system designed to reduce accidents caused by driver drowsiness by monitoring activity and providing warning alerts.



The background is a light beige color with various computer-related illustrations. In the top left, there are two green RAM modules. In the top right, there is a green GPU card with a white chip labeled 'GPU'. In the bottom left, there is a green motherboard with various ports and components. In the bottom right, there is a grey CPU chip labeled 'CPU'. There are also several orange circuit-like icons scattered around the text.

Welcome to

Discussion Time

Thank
You

